



Hands-on workshop on building a Simple Low-Field MRI Scanner Training Package

Overview: This workshop is organized by the Low-field Laboratory at the McConnell Brain Imaging Center of the Montreal Neurological Institute in collaboration with the Imaging Without Borders ([IMAGINE](#)) Summer School. In this interactive 2-hour workshop you will learn the basics of brain magnetic resonance imaging (MRI) hardware. You will assemble a simple MRI scanner, and then measure and characterize its magnetic field using a 3D field-mapping robot that you will build at the workshop.

Prerequisites: No prior hardware experience is needed. Your curiosity and a willingness to learn is all that is required.

Code of Conduct: The workshop is a harassment-free and safe environment for all, regardless of race, cultural affiliation, gender, nationality, beliefs, age, or disability. All participants are expected to comply with this code of conduct, and violations of its standards may result in dismissal from the workshop at the discretion of the workshop organizers. Upholding the standards of behavior is mandatory to maintain the workshop's integrity and create a positive environment. All participants are expected to behave professionally and responsibly towards everyone during all activities of the workshop, regardless of the session.

Safety Instructions: The workshop provides a hands-on introduction to low-field MRI engineering through the assembly and testing of an open-source educational scanner. Participants will work with basic tools, strong neodymium magnets, and simple electrical components in a guided environment. The neodymium magnets can have strong magnetic fields and can cause pinching injuries, damage electronic devices, or be damaged if not handled with caution. You may also work with electrical connections, do not touch any electrical components or connections, unless instructed and under supervision.

Workshop Activities: Over the course of the workshop, you will engage in:

1. Provide a Photo Use Consent
2. Complete a Pre-training Survey to help improve the workshop for future participants
3. Review safety and code of conduct agreements.
4. Participate in a Lecture on the Fundamentals of MRI Hardware for low-field brain imaging.
5. Work together to assemble a low-field MRI scanner and an MRI field-mapping system.

6. Assess the field of the low-field MRI scanner you build to see if it provides a good basis to scan brain images at a very low magnetic field.
7. Complete a Post-training Survey to help improve the workshop for future participants.
8. Participate in a Group photo to commemorate this fun experience

In preparation to the workshop, kindly receive the following:

QR Codes:  [QR Codes_Polytechnique](#)

Parts Checklist:  [Polytech_Part checklist](#)

Training Materials for the Workshop

1. Introductory lecture: Basic of Low-field MRI by Guillermo - [Introductory lecture](#)
2. How to build a 3D field mapper by Sipan Hovesipan - [DIY Field mapping system](#)
3. Animation of assembly of the ERNIE Education MRI scanner - [Animation](#)

Additional Information on Low-Field MRI

Here is more information to help you explore and better understand low-field brain MRI. This information is not mandatory to participate in the workshop. They are provided for participants that are more curious and want to continue learning about low-field MRI engineering and use.

Introductory Videos

1. How to design Halbach array scanner rings with Inventor by Timon- [Ring Design](#)
2. Introductory lecture on MaRCOS by Marina- [MaRCOS](#)
3. Introduction to Low-Field MRI by Dr. Johnes- [Part 1](#), [Part 2](#)
4. The low-field MRI revolution by Andrew Webb- [LF-MRI revolution](#)
5. High (56T), Medium (7T), and Low-field (50mT) by Andrew Webb - [HF](#), [MF](#), [LF](#)
6. DIY MRI: How to build your own MRI scanner MR by Tom O'Reilly- [DIY MRI](#)
7. OSI² ONE: An open-source low-field MRI scanner - [OSI² ONE](#)
8. OSI² Build Workshop | PTB, Berlin, 22.09.2023- [OSI² Build](#)
9. MRI4all- [MRI together](#)

Relevant Publications

1. Open-source magnetic resonance imaging: Improving access, science, and education through global collaboration- [OSI2 publication](#)
2. Design, Characterisation and Performance of an Improved Portable and Sustainable Low-Field MRI System- [publication](#)
3. In vivo 3D brain and extremity MRI at 50 mT using a permanent magnet Halbach array- [publication](#)
4. Tackling SNR at low-field: a review of hardware approaches for point-of-care systems- [publication](#)
5. Low-field MRI: Clinical promises and challenges- [publication](#)
6. On-site construction of a point-of-care low-field MRI system in Africa- [publication](#)
7. MRI4all- [Publication](#)
8. A Hands-On Workshop for Constructing a Low-Field MRI System in Three Days- [paper](#)

9. ezyMRI: How to build an MRI machine from scratch -- Experience from a four-day hackathon- [publication](#)
10. Design of Sparse Halbach Magnet Arrays for Portable MRI Using a Genetic Algorithm- [publication](#)
11. In vivo 3D brain and extremity MRI at 50 mT using a permanent magnet Halbach array- [publication](#)
12. In-vivo imaging with a low-cost MRI scanner and cloud data processing in low-resource settings- [paper](#)
13. Portable magnetic resonance imaging of patients indoors, outdoors, and at home- [publication](#)
14. Elliptical Halbach magnet and gradient modules for low-field portable magnetic resonance imaging- [publication](#)
15. Use of 2.1 MHz MRI scanner for brain imaging and its preliminary results in stroke- [publication](#)
16. Electromagnetic Noise Characterization and Suppression in Low-Field MRI Systems- [publication](#)
17. Assessment of Brain Injury Using Portable, Low-Field Magnetic Resonance Imaging at the Bedside of Critically Ill Patients- [publication](#)
18. A low-cost and shielding-free ultra-low-field brain MRI scanner- [publication](#)
19. A Low-Field Portable MRI Scanner With an Elliptic-Bore Halbach Magnet- [publication](#)

Open source hardware libraries

1. Halbach Ring Magnetic Field Optimization- [git-hub](#)
2. Field mapper robot- [git-hub](#)
3. Imaging without borders- [git-hub](#)
4. ERNIE
5. Open Source Imaging Initiative- [OSI²](#)
6. MRI4all - [MRI4all](#)
7. Halbach MRI Designer - [github](#)
8. Delta-DIY-MRI- [git-hub](#)
9. LUMC - [git-hub](#)
10. OSII MRI- [git-hub](#)